L2.5 Curve Sketching

§3.5 — precalc + the derivatives table

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1. A cubic, already factored. $y = (x-2)^2(x+4)$

Run **DIAS**. Mark the *type* of each zero. Then expand, build the derivatives table on one number line, and sketch.

> **Stop.** What goes in the **A** row for a polynomial? It is not blank.

2. Now a radical. $f(x) = \sqrt[3]{x}\,(x-4)$

Run **DIAS**, then build the **derivatives table**, then sketch.

Write f in powers of x before you differentiate. Report every row, including the ones whose answer is "none" — a test that fails still gets reported.

3. Look at it first. $y = 1 + \frac{2}{x} + \frac{1}{x^2}$

Sketch it. *Before* you differentiate anything, write one sentence about what this function can and cannot do.

4. A rational with a slant asymptote. $y = \dfrac{x^2 - 2x + 4}{x - 2}$

Full DIAS, then the table, then the sketch — with the slant asymptote drawn *on* the same axes, not beside them.

> **Stop.** Long division first. What is left over, and where does it go?

5. **BOARDS** **One rational per group. Sketch it completely on your wall.**

Group 1 $\quad y = \dfrac{1}{(x-3)^2(x+2)^2}$

Group 2 $\quad y = \dfrac{3x^2-8x-3}{x^3-x}$

Group 3 $\quad y = \dfrac{(x+2)^2(x-5)}{(x-3)(x+1)(x+4)}$

Group 4 $\quad y = \dfrac{2x^3-5x^2-11x-4}{3x^3+11x^2+5x-3}$

Factor before you decide anything. When every wall is up, compare them: each of these four has a feature the other three do not.